Individual Assignment Report

1. Characterization of the Problem, the Data, and the User Tasks

The goal of this application is to help a user decide in which US state or community they might want to settle by exploring the Communities and Crime dataset. The application uses the file communities.csv, in which each row corresponds to a community, and the table contains 128 attributes in total. Among these variables, 103 columns have no missing values. Semantically, these complete attributes can be categorized into several aspects, such as identification and indexing, population size, crime and social risk, etc. This is therefore a multivariate decision problem because users will consider multiple aspects at the same time.

The main user tasks are therefore exploratory and comparative. First, the user needs to compare communities in a quantitative space to detect trends, trade-offs, and outliers. Second, the user needs to understand how promising communities are distributed across states. Third, the user needs to narrow the search to a small set of candidate communities and inspect them in more detail. Finally, because different users may value criteria differently, the application should support a preference-driven interpretation of the data instead of relying on a single fixed ranking.

To support these tasks, the interface provides operations for selecting two quantitative attributes, brushing multiple communities in a scatterplot, hovering and clicking hierarchy nodes, selecting all communities from a state, and comparing the selected candidates in a summary panel. A weighted livability score is also computed from several dimensions so that the user can adapt the ranking to their own priorities.

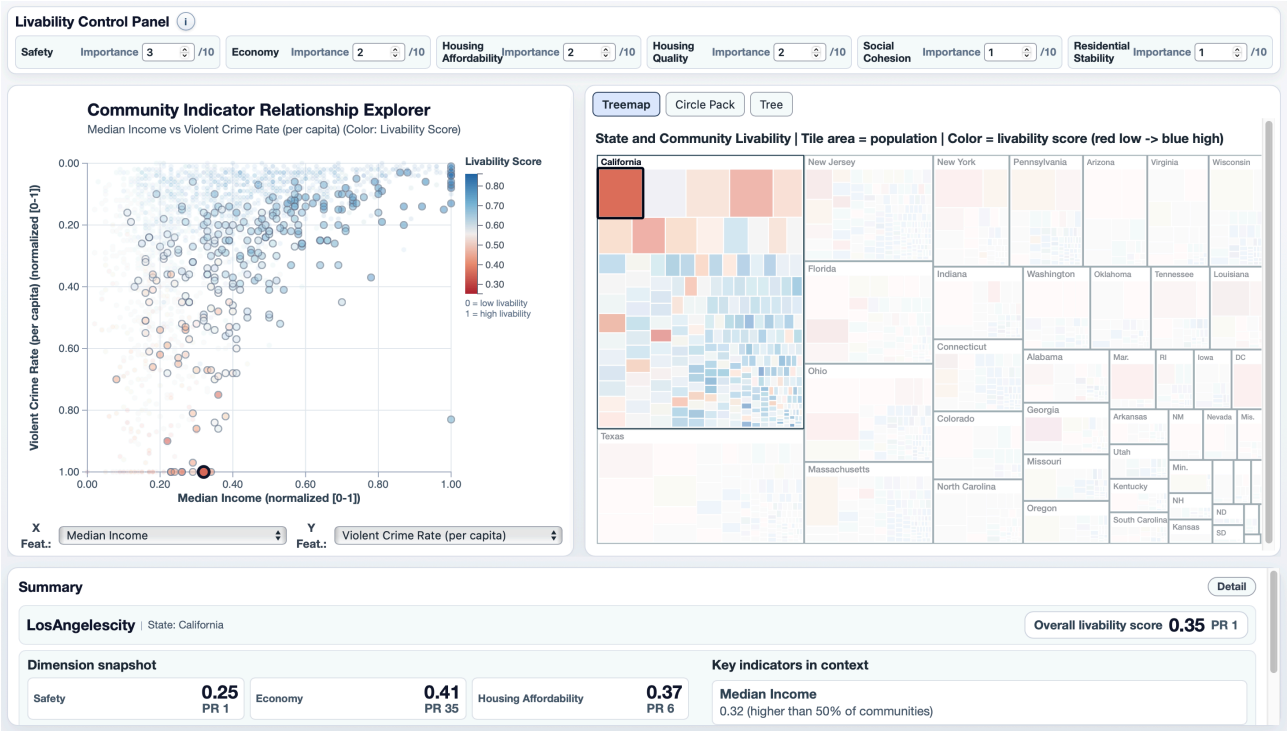
2. Design Rationale

The visual design is based on two coordinated views with complementary roles. The scatterplot is used to represent the relationship between two quantitative attributes chosen by the user. It is effective for identifying clusters, outliers and trade-offs, for example between income and violent crime. The color channel is used to encode the computed livability score, which lets the user see whether visually promising points in the quantitative space are also highly ranked overall. The 2D brush interaction is useful because it allows the user to select multiple candidate communities at once and send them to the linked views.

The hierarchy view organizes the data into two levels: state and community. This view complements the scatterplot because it emphasizes structure rather than correlation. It helps the user understand which states contain several promising communities and whether a brushed or selected subset is concentrated in a small number of states. Clicking a state selects all its communities, hovering a state highlights the corresponding points in the scatterplot, and hovering a community highlights the same item in both views. This synchronization reduces cognitive load. The summary panel then supports detailed inspection and small-set comparison after the main coordinated exploration step.

I tested three hierarchical layouts: treemap, circle packing, and tree. Treemap is the most space-efficient layout and provides the best overview when many communities must be displayed at once.

It also supports comparison because the state regions and community tiles remain compact and easy to scan, although labels become difficult to read for very small tiles. Circle packing produces a visually intuitive parent-child structure, but it uses space less efficiently. The tree layout makes the hierarchy explicit and easy to understand, but it becomes very long when many communities are shown, and it is less effective for overview tasks. My preferred layout is the treemap because it offers the best balance between readability, overview, density, and comparison, which matches the main analytical goal of shortlisting places from a large multivariate dataset.



3. Declaration of Generative AI Use

I first built the initial project structure and basic implementation by myself, based on the course tutorials. Then I used OpenAI Codex, a GPT-5-based coding assistant, to discuss design rationale, refine the coordinated interaction logic, debug state, and item synchronization between the scatterplot and the hierarchy, and help me implement complex functions and features that were not introduced in the course.

Some AI suggestions were rejected or adapted when they introduced unnecessary complexity. For example, AI suggested using a composite post-scoring index on the scatterplot X and Y axes. I rejected this idea because it would make it harder for users to inspect and compare detailed differences between communities. Instead, I chose to use raw features as the X and Y axis attributes so that the scatterplot remains interpretable and more useful for decision-making. The final implementation choices, the livability score and summary panel design, the selection of displayed attributes, the preferred hierarchy layout, the integration of the views, and testing were completed and validated by me.